Unsupervised Models Report

Unsupervised Models Report: Categorization of Radio Pulsar Signals

This report summarizes the performance, architecture, and results of the unsupervised machine learning models implemented for the categorization of radio pulsar signals. The three models evaluated are K-Means Clustering, Self-Organizing Map (SOM), and Principal Component Analysis (PCA).

All unsupervised models were trained and evaluated on raw (unscaled) features to closely match the methodology of the reference paper by Debesai, Gutierrez, and Koyluoglu.

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1. K-Means Clustering
K-Means clustering was applied to group the pulsars into distinct clusters based on their feature similarities.

Model Setup:
* Data: Raw, unscaled features.
* Optimal Clusters (k): Determined iteratively from k=2 to k=7 using Elbow Curve and Silhouette Scores.
* Target Clusters: Chosen as k=3 for the final execution.

Results:
* Silhouette Score: 0.4413 (Targeting ~0.44 as per paper).
* Within-Cluster Sum of Squares (WCSS) for k=3: 4,115,339.39

Cluster Distribution (k=3):
* Cluster 1: 974 samples | Centroid: [69.2, 39.2, 2.4, 10.6, 20.6, 48.5, 3.7, 18.8] | Target: ~974
* Cluster 2: 621 samples | Centroid: [34.4, 37.4, 4.4, 24.3, 99.1, 72.1, 0.4, -0.3] | Target: ~621
* Cluster 3: 44 samples | Centroid: [94.9, 45.2, 0.9, 2.4, 1.3, 12.8, 14.5, 255.6] | Target: ~44

Note: The model matched the paper's target distributions almost perfectly.

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2. Self-Organizing Map (SOM)
A SOM was implemented from scratch using NumPy to map the multidimensional dataset onto a lower-dimensional (2D) grid while preserving topology.

Model Setup:
* Architecture: 5x5 Output Grid (25 neurons).
* Epochs: 100 Iterations.
* Data: Raw, unscaled features.
* Macro-clustering: The 25 learned weight vectors were assigned to the nearest K-Means centroid (k=3) to

collapse them into 3 macro-clusters.

Results:
* Silhouette Score: 0.4024 (Targeting ~0.44).

Cluster Distribution (Macro-clusters):
* Cluster 1: 1,045 samples | Centroid: [70.1, 39.5, 2.3, 10.3, 20.9, 47.7, 4.1, 28.3] | Target: ~1,079
* Cluster 2: 594 samples | Centroid: [33.2, 37.3, 4.5, 24.8, 100.7, 71.9, 0.4, -0.3] | Target: ~517

Note: The 3rd target cluster (~43 samples) was absorbed during the neuron collapse in this specific execution, leaving 2 predominantly active macro-clusters.

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3. Principal Component Analysis (PCA)
PCA was utilized for dimensionality reduction and feature correlation analysis to understand which features drive the variance in the dataset.

Model Setup:
* Components: n = 2 Principal Components.
* Data: Raw, unscaled data.

Results:
* Explained Variance Ratio: PC1: 61.06%, PC2: 24.98%, Total Explained Variance: 86.04%

Feature Correlation Analysis (PCA Loadings):
* mean_integrated_profile : PC1 = 0.3175, PC2 = -0.2429
* std_integrated_profile : PC1 = 0.0327, PC2 = -0.0000
* kurtosis_integrated_profile : PC1 = -0.0186, PC2 = 0.0139
* skewness_integrated_profile : PC1 = -0.1194, PC2 = 0.0959
* mean_dmsnr_curve : PC1 = -0.5765, PC2 = 0.6630
* std_dmsnr_curve : PC1 = -0.2576, PC2 = 0.0578
* kurtosis_dmsnr_curve : PC1 = 0.0487, PC2 = 0.0083
* skewness_dmsnr_curve : PC1 = 0.6946, PC2 = 0.6991

Top contributor to PC1: skewness_dmsnr_curve
Top contributor to PC2: skewness_dmsnr_curve

(Because the unscaled skewness_dmsnr_curve and mean_dmsnr_curve have naturally massive variances, they heavily dominate the Principal Components.)